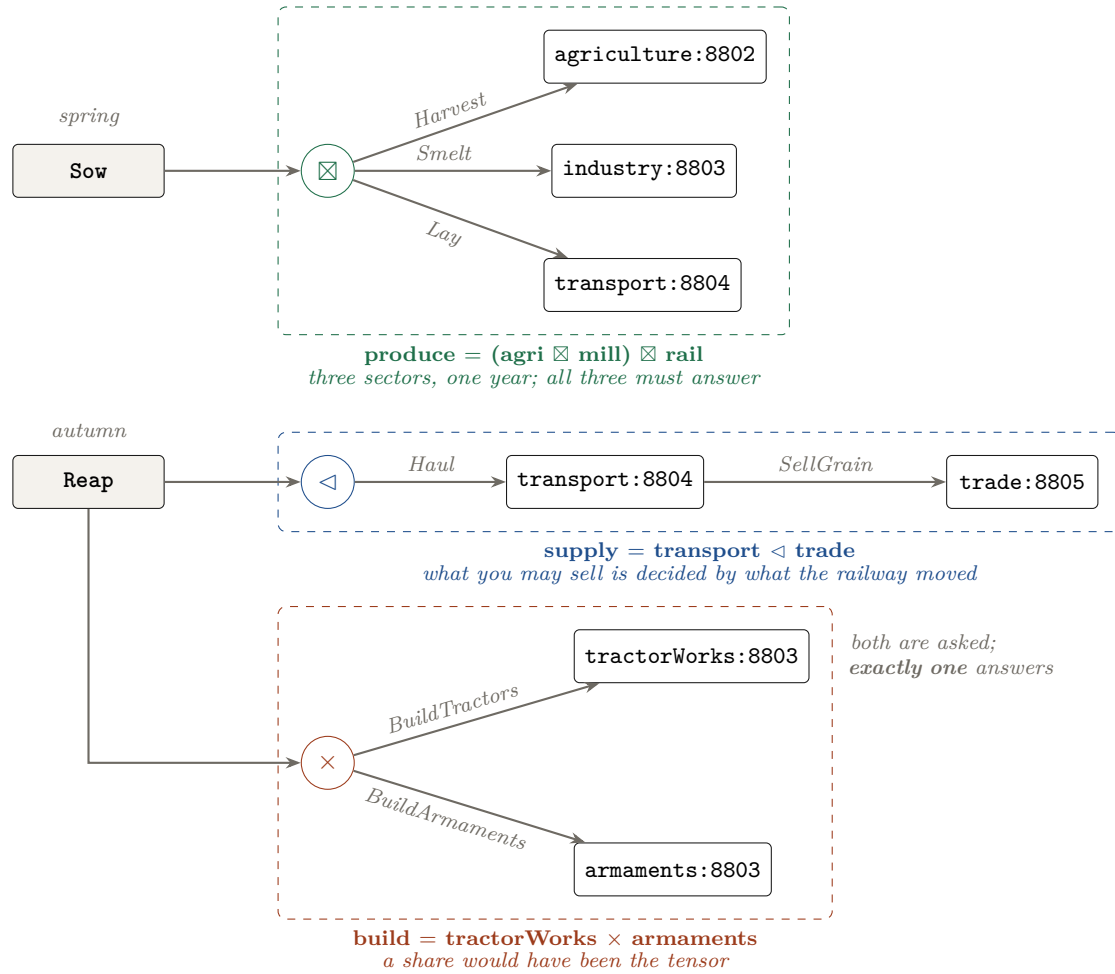


The four combinators, and where each one fires

stalin — §5 of Ghani, *Containers for Typed Agentic AI*

A container is (P, R) : a type of prompts, and a prompt-indexed type of valid replies.

Every box below is one. Every arrow is a prompt going down and a reply coming back.



what it means	where it fires in stalin
⊠ shapes multiply, positions multiply: ask both, and both must answer	the three sectors work the <i>same</i> year. both shapes are supplied <i>up front</i> , so neither can depend on the other's reply.
◁ the shape is a dependent pair ($s : A ** A.Pos s \rightarrow B$): a first prompt <i>together with</i> a function choosing the second from the first's reply	what can be <i>sold</i> depends on what the railway actually <i>moved</i> . The second prompt is not knowable when the first is sent.
× offer both, owe exactly one — Either (Rp) (Tq)	this year's steel becomes tractors <i>or</i> armaments. A <i>share</i> would have been the tensor; calling that a product would be false.
+	never built as a composite. It appears as each container's own shape type — a tagged union — and its eliminator is the switch in every server's dispatch. Real work, in the place §3 always put it; but no sumC (...) call, and it would be an overstatement to claim one.

Two that used to be here. While the line to the port was narrow the sale was a *product* (grain *or* steel, one contract) and became a *tensor* once enough rail was laid — rail capacity choosing between $\times$ and $\boxtimes$. Selling steel was removed from the game, so both would have been running on a quantity that is permanently zero. They were deleted rather than left for a diagram to claim.

One that broke and was repaired. Splitting the year put produce in spring and the haul in autumn, with a player decision between them, so **seqC**(produce, transport) could no longer run in a single call — it stayed declared and silently never fired. The dependent pair had not disappeared; it had moved one link down the chain. Hence **seqC**(transport, trade): haul, *then* sell what reached the port.